

SYNTHETIC PDF EXTRACTION STRESS-TEST DOCUMENT

Legal / Financial Technical Report — Benchmark Artifact

Document ID	DOC.04.7.19	Version	3.7
Generated	2026-08-28	Classification	Synthetic / Benchmark
Purpose	PDF parsing, OCR, layout and extraction evaluation	Pages	Dynamic
Record Type	Technical Review / Financial Record	Status	TEST ONLY

EXTRACTION WARNING

This document intentionally introduces competing reading regions, irregular spacing, synthetic encoding artifacts, and structurally complex tables.

Benchmark scope. The artifact is synthetic and contains no real transaction, person, account, or legal matter. It is designed to test whether a document-processing system can preserve semantics while dealing with difficult visual layouts.

Glyph sample: fi fl ffi x² H₂O ∑_{i=1}ⁿ x_i ∫₀¹ f(x) dx √(x²+y²) ±0.25 â€™™ â€œquoted textâ€œ Å© Å©” • ...

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1. Headers, Footers & Marginalia

MARGIN NOTE

NOTE A — Verify source order. This annotation is intentionally outside the primary narrative region.

The first review layer establishes a conventional legal-financial reading surface and then interrupts it with secondary annotations. The main narrative should be read top-to-bottom, while the margin notes, callouts, and footer notices form independent extraction regions. Reference [1] is deliberately placed in running prose.

REVIEWER NOTE

Values in this section are synthetic. Do not interpret them as evidence of an actual transaction.

Invoice ID: 98402
Customer Code: CX-771
Processing Status: Pending Review

A borderless alignment test follows. It relies entirely on whitespace and does not use cell borders.

Invoice ID	98402	Currency	USD
Customer Code	CX-771	Status	Pending
Subtotal	12,480.00	Tax	1,497.60
Total	13,977.60	Due Date	2026-09-15
Reference No.	REF-2026-0048	Risk	Medium

Footnote [1] appears at the bottom of this page in the rendered document. A longer sentence is used so that the footnote itself wraps and remains visually distinct from the body.

1. Synthetic reference used to test footnote separation from body text; the wording intentionally continues long enough to wrap across multiple lines and challenge reading-order reconstruction.

2. Operational Findings

The extraction pipeline must preserve the logical reading order even when the visual layout introduces competing text regions.

EXTRACTION WARNING

This callout sits between logical paragraphs and is intended to interrupt visual flow.

The first column contains narrative findings, reviewer observations, and artificially broken words such as imple-
mentation, docu-
ment, extra-
ction, and multi-
column. These line fragments are intentional and should not be silently normalized when evaluating raw extraction fidelity.

The system should distinguish a genuine line break from a paragraph boundary. It should also preserve punctuation, unusual whitespace, and inline references such as [2].

Finding 2.1. The body continues beneath a secondary annotation region. The annotation itself is not part of the paragraph.

MARGIN NOTE

OCR CHECK — compare O/O
and I/I/I.

A synthetic identifier such as TXN_98402_07 appears in the prose, followed by a URL-like token <https://example.invalid/ref/98402> and a mailbox-like string review@example.invalid.

Additional prose is included to create realistic density. The benchmark should reward systems that identify semantic continuity rather than simply reading objects in x/y coordinate order.

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2.2 Continued Review

Column two contains a continuation of the operational narrative. The sidebar is visually adjacent but logically independent. A small list is intentionally interleaved with prose.

- First observation: whitespace alignment can encode table structure.
- Second observation: callouts can create competing reading paths.
- Third observation: repeated headers must not be mistaken for body content.

The next paragraph uses multiple typographic voices on a single line: Bold normal italic monospace bold italic SMALL CAPS.

Encoding sample: â€™ â€œquoted textâ€ Â© Â± â€”
â€” • ... and clean equivalents: “quoted text”, ©, ±, –
— • ...

Metric	Value	Reviewer
Latency	184 ms	PASS
Recall	97.4%	CHECK
Variance	-0.84	NOTE

The conclusion returns to a full-width reading context on the following page. This deliberate frame transition tests whether the extraction engine understands page-level layout changes.

3. Three-Column Review

Column 1: narrative text. The report describes a synthetic review cycle and a sequence of document-handling decisions. The prose is dense enough to make column ordering meaningful.

Column 1 continuation: the record contains identifiers, dates, and short conclusions. A vertical label on the page edge is separate from this content.

Column 2: statistics and short paragraphs. Values include 1,000; 25,000.75; -842.10; 3.14159; ±0.25. Percentages should remain numeric.

STATISTICS

N = 48; rate = 7.25%; variance = -0.84; status = REVIEW.

Column 2 continuation: mathematical glyphs Σ $\int$ $\sqrt{}$ $\leq$ $\geq$ $\neq$ $\approx$ Δ π appear in prose and formulas.

Column 3: reviewer observations. The reviewer notes that a visual adjacency relationship does not imply semantic inclusion.

MARGIN NOTE

Reviewer comment: table continues.

The final observation explicitly references §4.2 and NOTE A. This cross-reference is synthetic and intentionally short.

A tiny checklist follows:
[X] Approved
[] Rejected
[X] Requires Review
[] Escalated

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4. Table Extraction Complexity

The next table intentionally spans multiple pages. Its header repeats after page breaks. It contains null cells, N/A markers, negative values, percentages, dates, and mixed alignment.

Item	Description	Category	Qty	Unit Price	Discount	Tax	Net	Status	Review Date
ITM-001	Synthetic transaction record 01 with intentional nulls and long descriptive text for row-boundary testing.	Category A	10	\$11.94	2.5%	2.00%	119.43	Pending	2026-02-02
ITM-002	Synthetic transaction record 02 with intentional nulls and long descriptive text for row-boundary testing.	Category B	15	\$15.91	5.0%	3.00%	238.86	Approved	2026-03-03
ITM-003	Synthetic transaction record 03 with intentional nulls and long descriptive text for row-boundary testing.	Category C	20	\$19.15	7.5%	4.00%	358.29	N/A	2026-04-04
ITM-004	Synthetic transaction record 04 with intentional nulls and long descriptive text for row-boundary testing.	Category D	30	\$15.91	10.0%	5.00%	477.72	Open	2026-05-05
ITM-005	Synthetic transaction record 05 with intentional nulls and long descriptive text for row-boundary testing.	Category E	40	\$14.66	12.5%	6.00%	597.15	Pending	2026-06-06
ITM-006	Synthetic transaction record 06 with intentional nulls and long descriptive text for row-boundary testing.	Category F	50	\$14.32	15.0%	7.00%	716.58	Approved	2026-07-07
ITM-007	Synthetic transaction record 07 with intentional nulls and long descriptive text for row-boundary testing.	Category G	60	\$14.34	0.0%	8.00%	836.01	N/A	2026-08-08
ITM-008	Synthetic transaction record 08 with intentional nulls and long descriptive text for row-boundary testing.	Category H	70	\$13.50	2.5%	9.00%	955.44	Open	2026-09-09
ITM-009	Synthetic transaction record 09 with intentional nulls and long descriptive text for row-boundary testing.	Category I	80	\$13.36	5.0%	1.00%	1,074.87	Pending	2026-01-10
ITM-010	Synthetic transaction record 10 with intentional nulls and long descriptive text for row-boundary testing.	Category J	90	\$12.22	7.5%	2.00%	1,194.30	Approved	N/A
ITM-011	Synthetic transaction record 11 with intentional nulls and long descriptive text for row-boundary testing.	Category K	100	\$10.13	10.0%	3.00%	1,013.73	N/A	2026-03-12
ITM-012	Synthetic transaction record 12 with intentional nulls and long descriptive text for row-boundary testing.	Category L	110	\$12.13	12.5%	4.00%	1,433.16	Open	2026-04-13
ITM-013	Synthetic transaction record 13 with intentional nulls and long descriptive text for row-boundary testing.	Category M	120	\$12.96	15.0%	5.00%	1,552.59	Pending	2026-05-14
ITM-014	Synthetic transaction record 14 with intentional nulls and long descriptive text for row-boundary testing.	Category N	130	\$12.87	0.0%	6.00%	1,672.02	Approved	2026-06-15
ITM-015	Synthetic transaction record 15 with intentional nulls and long descriptive text for row-boundary testing.	Category O	140	\$13.69	2.5%	7.00%	1,791.45	N/A	2026-07-16
ITM-016	Synthetic transaction record 16 with intentional nulls and long descriptive text for row-boundary testing.	Category P	150	\$12.72	5.0%	8.00%	1,910.88	Open	2026-08-17
ITM-017	Synthetic transaction record 17 with intentional nulls and long descriptive text for row-boundary testing.	Category Q	160	\$12.50	7.5%	9.00%	2,030.31	Pending	2026-09-18
ITM-018	Synthetic transaction record 18 with intentional nulls and long descriptive text for row-boundary testing.	Category R	170	\$12.91	10.0%	1.00%	2,149.74	Approved	2026-01-19
ITM-019	Synthetic transaction record 19 with intentional nulls and long descriptive text for row-boundary testing.	Category S	180	\$12.25	12.5%	2.00%	2,269.17	N/A	2026-02-20
ITM-020	Synthetic transaction record 20 with intentional nulls and long descriptive text for row-boundary testing.	Category T	190	\$12.56	15.0%	3.00%	2,388.60	Open	N/A
ITM-021	Synthetic transaction record 21 with intentional nulls and long descriptive text for row-boundary testing.	Category U	200	\$12.91	0.0%	4.00%	2,508.03	Pending	2026-04-22
ITM-022	Synthetic transaction record 22 with intentional nulls and long descriptive text for row-boundary testing.	Category V	210	\$11.30	2.5%	5.00%	2,327.46	Approved	2026-05-23
ITM-023	Synthetic transaction record 23 with intentional nulls and long descriptive text for row-boundary testing.	Category W	220	\$12.56	5.0%	6.00%	2,746.89	N/A	2026-06-24
ITM-024	Synthetic transaction record 24 with intentional nulls and long descriptive text for row-boundary testing.	Category X	230	\$12.46	7.5%	7.00%	2,866.32	Open	2026-07-25
ITM-025	Synthetic transaction record 25 with intentional nulls and long descriptive text for row-boundary testing.	Category Y	240	\$12.15	10.0%	8.00%	2,985.75	Pending	2026-08-26
ITM-026	Synthetic transaction record 26 with intentional nulls and long descriptive text for row-boundary testing.	Category Z	250	\$12.03	12.5%	9.00%	3,105.18	Approved	2026-09-27
ITM-027	Synthetic transaction record 27 with intentional nulls and long descriptive text for row-boundary testing.	Category AA	260	\$12.00	15.0%	1.00%	3,224.61	N/A	2026-01-01
ITM-028	Synthetic transaction record 28 with intentional nulls and long descriptive text for row-boundary testing.	Category AB	270	\$12.76	0.0%	2.00%	3,344.04	Open	2026-02-02
ITM-029	Synthetic transaction record 29 with intentional nulls and long descriptive text for row-boundary testing.	Category AC	280	\$12.69	2.5%	3.00%	3,463.47	Pending	2026-03-03
ITM-030	Synthetic transaction record 30 with intentional nulls and long descriptive text for row-boundary testing.	Category AD	290	\$12.50	5.0%	4.00%	3,582.90	Approved	N/A
ITM-031	Synthetic transaction record 31 with intentional nulls and long descriptive text for row-boundary testing.	Category AE	300	\$12.41	7.5%	5.00%	3,702.33	N/A	2026-05-05
ITM-032	Synthetic transaction record 32 with intentional nulls and long descriptive text for row-boundary testing.	Category AF	310	\$12.00	10.0%	6.00%	3,821.76	Open	2026-06-06
ITM-033	Synthetic transaction record 33 with intentional nulls and long descriptive text for row-boundary testing.	Category AG	320	\$11.25	12.5%	7.00%	3,641.19	Pending	2026-07-07
ITM-034	Synthetic transaction record 34 with intentional nulls and long descriptive text for row-boundary testing.	Category AH	330	\$12.66	15.0%	8.00%	4,060.62	Approved	2026-08-08
ITM-035	Synthetic transaction record 35 with intentional nulls and long descriptive text for row-boundary testing.	Category AI	340	\$11.44	0.0%	9.00%	4,180.05	N/A	2026-09-09
ITM-036	Synthetic transaction record 36 with intentional nulls and long descriptive text for row-boundary testing.	Category AJ	350	\$11.40	2.5%	1.00%	4,299.48	Open	2026-01-10
ITM-037	Synthetic transaction record 37 with intentional nulls and long descriptive text for row-boundary testing.	Category AK	360	\$12.47	5.0%	2.00%	4,418.91	Pending	2026-02-11
ITM-038	Synthetic transaction record 38 with intentional nulls and long descriptive text for row-boundary testing.	Category AL	370	\$12.25	7.5%	3.00%	4,538.34	Approved	2026-03-12
ITM-039	Synthetic transaction record 39 with intentional nulls and long descriptive text for row-boundary testing.	Category AM	380	\$11.75	10.0%	4.00%	4,657.77	N/A	2026-04-13
ITM-040	Synthetic transaction record 40 with intentional nulls and long descriptive text for row-boundary testing.	Category AN	390	\$11.90	12.5%	5.00%	4,777.20	Open	N/A
ITM-041	Synthetic transaction record 41 with intentional nulls and long descriptive text for row-boundary testing.	Category AO	400	\$12.17	15.0%	6.00%	4,896.63	Pending	2026-06-15
ITM-042	Synthetic transaction record 42 with intentional nulls and long descriptive text for row-boundary testing.	Category AP	410	\$12.35	0.0%	7.00%	5,016.06	Approved	2026-07-16
ITM-043	Synthetic transaction record 43 with intentional nulls and long descriptive text for row-boundary testing.	Category AQ	420	\$12.73	2.5%	8.00%	5,135.49	N/A	2026-08-17
ITM-044	Synthetic transaction record 44 with intentional nulls and long descriptive text for row-boundary testing.	Category AR	430	\$12.44	5.0%	9.00%	4,954.92	Open	2026-09-18
ITM-045	Synthetic transaction record 45 with intentional nulls and long descriptive text for row-boundary testing.	Category AS	440	\$11.67	7.5%	1.00%	5,374.35	Pending	2026-01-19

4.4 Merged Cells and Nested Table

The following structure combines colspan and rowspan behavior with an actual nested table contained inside a parent cell.

FINANCIAL DATA				
Quarter	Q1	Q2	Q3	Q4
Actual	10,000	12,500		18,000
	11,000	13,100	16,750	19,200
	-1,000	-600	—	±0.25
Nested Details				

Additional table edge cases follow: empty cells, a cell containing only —, a zero, N/A, and a long unbroken identifier
DOC.04.7.19/CASE-26/771/REF-2026-0048-A.

Nested lists inside a table-adjacent narrative:

- 1
- Primary extraction path
- 2
- Secondary review path

5. Typography, Encodings & Glyphs

Ligature candidates: financial • official • efficient • affiliation • flexible • offline.

Bold normal italic monospace bold italic SMALL CAPS

Superscripts and subscripts: x^2 , x^3 , H_2O , CO_2 , $E = mc^2$, $a_1 + a_2 + a_3$.

Mathematics: $\sum_{i=1}^n x_i$ $\int_0^1 f(x) \, dx$ $\sqrt{(x^2 + y^2)}$ $\Delta P = P_2 - P_1$

Encoding artifact panel

â€™ â€œquoted textâ€ Â© Â± â€” â€” • ... “clean quote” © ±

Unusual whitespace and punctuation are intentionally mixed. The next line contains a long token:
REF-2026-0048-A__TXN_98402_07__CASE/26/771__DOC.04.7.19

SECTION 04

ROTATED TEXT

A separate page-edge label is rendered vertically to test extraction of rotated text objects.

90° and 270° labels should be interpreted as text objects even though their visual baseline is vertical. The document also contains serif and sans-serif sections to test font identification.

6. Synthetic Form / Record Extraction

Key-value pairs deliberately use dots, underscores, inconsistent spacing, and mixed alignment.

Invoice ID:		98402
Customer ID:	_____	CX-771
Reference No.:		REF-2026-0048
Account Type:		Enterprise
Processing Status:		Pending Review
Amount Due:		\$13,977.60
Review Date:		2026-09-15

Status selection

☒ Approved ☐ Rejected ☒ Requires Review ☐ Escalated ☐ Archived

Schedule

☒ Monthly ☐ Quarterly ☐ Annual

Priority

☐ Low ☒ Medium ☐ High

Signature blocks

Authorized Representative: _____

Signature: _____

Date: _____

Reviewer: _____

Signature: _____

REVIEWER NOTE
One signature block is intentionally close to the bottom of the page to test boundary detection.

7. Reading-Order Stress Test

MARGIN NOTE
IMPORTANT — sidebar content is not part of the paragraph immediately beside it.

VISUAL ORDER CONFLICT
The sidebar begins higher than the body paragraph that logically precedes it.

Primary narrative. Full-width heading appears first. Main body begins in the left region. A sidebar begins higher on the page. The second narrative block begins before the sidebar ends. This is intentional.

SIDEBAR	CONTENT
A	Reviewer observation: independent text region.
B	Short metric: 97.4%
C	Status: Pending

The second narrative block continues below the sidebar. A reading-order engine should not splice the sidebar rows into the body sentence merely because their bounding boxes overlap vertically.

Footnote [2]. This note is positioned low on the page and intentionally resembles a separate document object. It should be associated with the superscript-like reference in the narrative.

2. This note intentionally contains a longer sentence that wraps across multiple lines and creates a competing low-page reading region.

8. OCR / Visual Noise Section

The following block imitates a poorly digitized synthetic record while retaining enough clarity to remain usable.

INV0ICE ID: 98402

CUSTOMER C0DE: CX-77I

TOTAL: \$13,977.60

Mixed capitalization: Pending REVIEW / pending review / PENDING REVIEW.

Irregular spacing: C U S T O M E R C O D E C X - 7 7 1

Missing-space sample: ProcessingStatusPendingReview

Ambiguous characters: O/0, I/1/I, S/5, B/8, Z/2.

OCR CHECK

Compare visually similar characters and punctuation. The content remains synthetic and should not be treated as real data.

A long unbroken string follows for tokenization testing:

ABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789REF20260048TXN9840207CASE26771D0C04719

The remainder of the page contains ordinary text to ensure that OCR noise is localized rather than global. A benchmark should distinguish genuine text errors from intentional synthetic corruption.

9. Deliberate Edge Cases

Short paragraph.

N/A

0

—

Field	Value	Notes
Empty		intentionally null
Dash	—	single glyph
Zero	0	numeric zero
Not applicable	N/A	text marker
Long cell	A very long table-cell value that wraps across several lines to test wrapping boundaries and preservation of complete	

Nested list:

- 1
- 2
- Numbered item one

Numbered item two

Nested bullet A

Nested bullet B

A quotation contains brackets and punctuation: “Review [Section 04] before signing.” Another line contains (parentheses), {braces}, and tokens.

Dates appear as 2026-09-15, 15/09/2026, September 15, 2026, and 09.15.2026. Currency symbols: \$, €, £, ¥. Percentages: 7.25%, 100%, -0.84%.

10. Extraction Test Checklist

The final page summarizes the structures deliberately represented in the document. These labels are themselves extraction targets.

☒	Running headers
☐	Running footers
☐	Page X of Y
☒	Sidenotes
☐	Footnotes
☐	Two-column layout
☒	Three-column layout
☐	Floating callouts
☐	Borderless table
☒	Multi-page table
☐	Merged cells
☐	Nested table
☒	Ligatures
☐	Superscripts/subscripts
☐	Mathematical symbols
☒	Encoding artifacts
☐	Rotated text
☐	Key-value pairs
☒	Checkboxes
☐	Radio buttons
☐	Signature blocks
☒	Reading-order conflicts
☐	OCR noise

END OF SYNTHETIC DOCUMENT — EXTRACTION STRESS TEST